

NORMAL INCIDENCE — Γ AND τ

Med. 1 ($z < 0$): incident side. **Med. 2** ($z \geq 0$): transmitted side. Boundary at $z = 0$.

Reflection coeff. *wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{unitless})$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Intrinsic impedance *general* $\rightarrow$ *nonmag. shortcut*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default $\mu_r = 1$ (air, dielectric, “nonmag.”). Only use $\mu_r \neq 1$ if problem says so (“ferromagnetic”, iron, ferrite, or μ_r given explicitly).

Low-loss η_2 (poor conductor) $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon}\right) \xrightarrow{\text{drop tiny } j} \frac{\eta_0}{\sqrt{\epsilon_{r,2}}} \quad (\Omega)$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_{r,2}}$. The j term is for accuracy with $|\eta_c|, \theta_\eta$.

$\hat{E}$ PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \quad (\text{rad/m}) \quad \Rightarrow \quad k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp jk_1 u}, \quad \hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm jk_1 u}, \quad \hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp jk_2 u}$$

(lossy med 2: replace $e^{\mp jk_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j\beta_2 u}$)

Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$, $\epsilon_0 \approx 1/(36\pi \times 10^9)$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	$\sigma/\omega\epsilon$	η_c , α , β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact; $\alpha = 0$, $\beta = \omega\sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon_r}\right)$; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	$10^{-2} - 10^2$	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$, see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$; $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$

For Γ/τ : drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \quad \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon_r}$$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4*

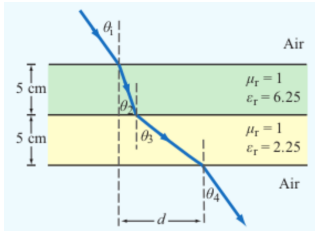
$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i , refracted angle inside slab i

Slab-indexed $\theta_i = \text{angle inside slab } i$

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Angle-indexed (HW3 P3 style) $\theta_1 = \text{incident}$, $\theta_{i+1} = \text{inside slab } i$

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes $t_1 \tan \theta_2$, slab 2 contributes $t_2 \tan \theta_3$. The angle in the formula is always the one the ray makes inside that slab.

Air–slab–air: exit angle returns to θ_1 but is laterally shifted by d .

FRESNEL OBLIQUE — Γ, τ, R, T

Perpendicular ($E \perp$ plane of inc.)

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad \tau_{\perp} = 1 + \Gamma_{\perp}$$

Parallel ($E \parallel$ plane of inc.)

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad \tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$$

Critical angle (TIR) *only if $n_1 > n_2$*

$$\sin \theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \Rightarrow \theta_i \geq \theta_c = \text{total internal refl.}$$

Brewster angle ($\parallel$ only) $\Gamma_{\parallel} = 0$, *no reflection*

$$\tan \theta_B = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \quad (\text{nonmag.})$$

Reflectivity / Transmissivity *power form; $R + T = 1$*

$$R = |\Gamma|^2, \quad T = |\tau|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$$

Normal incidence: $\theta_t = \theta_i = 0$, *cos factors drop*, $R = |\Gamma|^2$, $T = |\tau|^2 (\eta_1/\eta_2)$.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Shorthand: $c_i = \cos \theta_i$, $c_t = \cos \theta_t$.			
	Normal ($\theta_i = 0$)	$\perp$ pol	$\parallel$ pol
Γ	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$	$\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$	$\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{c_i}{c_t}$
R	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$
T	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\perp} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$	$ \tau_{\parallel} ^2 \frac{\eta_2 c_i}{\eta_1 c_t}$
$R+T$	$1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; $\eta_i = \sqrt{\mu_i / \epsilon_i}$; nonmag. $\Rightarrow \eta_2/\eta_1 = n_1/n_2$.

FIBER OPTICS

Core index n_f , cladding n_c , surrounded by n_0 . Need $n_f > n_c$ for TIR.

Index of refraction *nonmag.* $\mu_r = 1$

$$n = c/u_p = \sqrt{\epsilon_r}$$

Acceptance angle θ_a *max θ_i for TIR in fiber*

$$\sin \theta_a = \frac{1}{n_0} \sqrt{n_f^2 - n_c^2}$$

Modal dispersion (max time delay) *fiber length ℓ*

$$\tau = \frac{\ell n_f}{c} \cdot \frac{n_f - n_c}{n_c}$$

Max data rate $T \geq 2\tau$ *to avoid pulse overlap*

$$f_p = \frac{1}{2\tau} = \frac{c n_c}{2\ell n_f (n_f - n_c)}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0$, $H_z \neq 0$

TM mode $H_z = 0$, $E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE_{mn} vs TM_{mn} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” $\rightarrow$ TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).

3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE_{10} exists but TM_{10} doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: $C_X = \cos \frac{m\pi x}{a}$, $S_X = \sin \frac{m\pi x}{a}$, $C_Y = \cos \frac{n\pi y}{b}$, $S_Y = \sin \frac{n\pi y}{b}$. All fields carry an $e^{-j\beta z}$ factor (omitted below). $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$.

TE mode *generator: $\hat{H}_z$*

$$\hat{H}_z = H_0 C_X C_Y$$

$$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$$

$$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TE}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TE}}$$

TM mode *generator: $\hat{E}_z$*

$$\hat{E}_z = E_0 S_X S_Y$$

$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$$

$$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TM}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference: $\hat{E}_x = E_0 e^{-j\beta z}$, $\hat{H}_y = \hat{E}_x/\eta$, f_c N/A, $k = \omega\sqrt{\mu\epsilon}$.

CUTOFF FREQUENCY

f_{mn} m, n integers ≥ 0 (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad (\text{Hz})$$

Common cutoffs ($a > b$)

Quantity	Formula
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE_{10}	$u_{p0}/(2a)$ (dominant)
TE_{01}	$u_{p0}/(2b)$
TE_{20}	$u_{p0}/a = 2f_{10}$
$\text{TE}_{11}, \text{TM}_{11}$	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ϵ_r FROM DISPERSION

Given ω , β , mode (m, n)

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \quad (\Omega) \quad Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{\text{TE}}} \quad (\text{TE mode})$$

$Z_{\text{TE}} > \eta$ *always* (denominator < 1); $Z_{\text{TM}} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \quad (\text{m/s})$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \quad (\text{m/s}) \quad \Rightarrow \quad u_p u_g = u_{p0}^2 \quad (\text{m}^2/\text{s}^2)$$

$$t = L/u_g \quad (\text{s}) \quad \text{travel time through guide of length } L$$

Higher modes have lower $u_g \Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η_1, η_2 — imp. (Ω)
 Γ — refl. coeff (unitless)
 $\tau = 1 + \Gamma$ — trans. coeff
 $k_i = \omega \sqrt{\epsilon_{r,i}} / c$ (rad/m)
 α_2, β_2 — if med 2 lossy
 $z = 0$ — boundary plane
 $\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
 $n_i = \sqrt{\epsilon_{r,i}}$ — index of refr.
 t_i — slab thickness (m)
 d — lateral disp. (m)
plane of inc. = plane of $\hat{k}$ +normal
 $\perp$: $\mathbf{E} \perp$ this plane
 $\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
 $a > b$
 m, n — mode indices (int.)
 f_{mn} — cutoff (Hz)
 β — prop. const (rad/m)
 $Z_{\text{TE/TM}}$ — wave imp. (Ω)
 E_x, H_y — field comps (V/m, A/m)
 $\eta = \eta_0 / \sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c / \sqrt{\epsilon_r}$ — bulk phase vel.
 $u_p = u_{p0} / \sqrt{1 - (f_c/f)^2}$
 $u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$
 $u_p u_g = u_{p0}^2$
 $t = L / u_g$ — travel time
 $c = 3 \times 10^8$ m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$
 $\%P_t = 100|\tau|^2 \eta_1 / \eta_2$
 $\%P_r + \%P_t = 100$
 σ (S/m), ϵ (F/m), μ (H/m)
 $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
 $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $1 / (36\pi) \times 10^{-9} \approx \epsilon_0$

DIPOLE TYPE BY LENGTH

Check first <i>compute $\lambda = c/f$, then l/λ</i>		
l/λ	Type	Use
$< 1/50$	Hertzian (short)	Hertzian formula
$= 1/4$	Quarter-wave	Arbitrary formula
$= 1/2$	Half-wave	$D = 1.64$, simple
other	Arbitrary	Arbitrary formula

If $l/\lambda < 1/50$ stop and use Hertzian only — don't subst. small l into arbitrary formula.

HERTZIAN DIPOLE — $S(R, \theta)$

Power density <i>$l/\lambda < 1/50$; far field</i>
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32 \pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$
$k = 2\pi/\lambda \text{ rad/m; } I_0 \text{ peak current (A); } l \text{ length (m); } R \text{ obs. dist (m).}$
Max direction $\theta = \pi/2$ (broadside)
Total radiated power <i>integrate over $4\pi \text{ sr}$</i>
$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
Directivity <i>Hertzian, exact</i>
$D_{\text{Hertz}} = 1.5 \text{ (i.e. 1.76 dB)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at distance R <i>any l</i>
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
Normalized pattern $F(\theta) = S(\theta)/S_{\text{max}}$, <i>dimensionless</i>
Half-wave $l = \lambda/2 \quad \pi l/\lambda = \pi/2$
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$
Quarter-wave $l = \lambda/4 \quad \pi l/\lambda = \pi/4, \cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2} \right)^2 \approx 0.0858 S_0$

BEAM PARAMETERS — HPBW, Ω_p, D

1. Normalize
$F(\theta) = S(\theta)/S_{\text{max}}$
2. Half-power beamwidth (HPBW) <i>solve $F(\theta_{1/2}) = 0.5$</i>
$\beta = 2\theta_{1/2} \text{ (symmetric beams)}$
example $F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186 \text{ rad} \Rightarrow \beta = 21.31^\circ$
3. Pattern solid angle <i>(sr)</i>
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
If F has no ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$
4. Directivity
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
5. Gain $\xi = \text{radiation efficiency, } 0 < \xi \leq 1$
$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
<i>For lossless / ideal antennas: $G = D$.</i>

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

EFFECTIVE AREA A_e

Definition m^2 , <i>antenna's receiving cross section</i>
$A_e = \frac{\lambda^2 D}{4\pi}$
Physical cross section <i>half-wave dipole, wire diameter d</i>
$A_p = l d = \frac{\lambda}{2} d$
Ratio $A_e \gg A_p$ <i>for thin wires (good!)</i>
Example: $f = 100 \text{ MHz}, d = 2 \text{ cm} \Rightarrow A_e/A_p \approx 39$.

POWER FLOW IDENTITIES

$P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D}$
$S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$
<i>For isotropic radiator: $S = P_{\text{rad}}/(4\pi R^2)$.</i>

FRIIS TRANSMISSION FORMULA

Far-field link budget $P_t \text{ tx, } P_r \text{ rx, dist } R$
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$
G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$
Solve for G_r given A_e
$G_r = \frac{4\pi A_{e,r}}{\lambda^2} \Rightarrow P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$
<i>Use linear G, not dB. Convert first: $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$.</i>
<i>Free-space path loss $L_{\text{fs}} = (4\pi R/\lambda)^2$ so $P_r = G_t G_r P_t / L_{\text{fs}}$.</i>

dB $\leftrightarrow$ LINEAR

Power gain <i>factor $\rightarrow$ dB</i>
$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$
dB linear dB linear
0 1 10 10
3 2 13 20
6 4 20 100
7 5 30 1000
−3 0.5 −10 0.1
<i>For E or amplitude use $20 \log$ not $10 \log$.</i>

RADAR EQUATION (BONUS)

Monostatic radar $\text{tx} = \text{rx, target cross section } \sigma_t \text{ (m}^2\text{)}$
$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$
Doppler shift <i>closing velocity v_r</i>
$f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c} \text{ (Hz)}$
<i>Factor of 2 from round-trip path; v_r is the radial component toward radar.</i>

LECTURE-11 BOUNDARY $\leftrightarrow$ T-LINE

EM	T-line	Match
$\tilde{E}_i$	$\tilde{V}_1^+$	forward wave
$\tilde{E}_r$	$\tilde{V}_1^-$	reverse wave
η_i	Z_{0i}	line impedance
Γ	Γ	same formula
k_i	β_i	wavenumber

Bouncing-wave analogy: every t-line formula in Ch. 2 transfers verbatim with the substitutions above.

ANTENNA INTEGRATION TIPS

<i>Many radiation integrals (e.g. Ω_p for $e^{-20\theta^2} \sin \theta$) have no closed form.</i>
<i>Use Wolfram / numerical integration. State this in the answer.</i>
<i>Common shortcut: if $F(\theta)$ is narrow ($\theta_{1/2} \ll 1$), approximate</i>
$\Omega_p \approx \theta_{1/2}^2 \pi / (4 \ln 2) \text{ (Gaussian beam)}$

DIPOLE

l — antenna length (m)
$\lambda = c/f$ — wavelength (m)
I_0 — peak current (A)
$\eta_0 = 120\pi \approx 377 \Omega$
$k = 2\pi/\lambda$ — wavenumber (rad/m)
R_{rad} — radiation res. (Ω)
R — obs. dist (m)

RADIATION

$S(R, \theta)$ — power density (W/m^2)
S_{max} — peak power density
$F(\theta) = S/S_{\text{max}}$ — normalized
Ω_p — pattern solid angle (sr)
β — HPBW (rad or deg)
D — directivity (unitless)
P_{rad} — total rad. power (W)

ANTENNA

$A_e = \lambda^2 D / (4\pi)$ — effective area
A_p — physical cross sec. (m^2)
ξ — rad. efficiency ($0 < \xi \leq 1$)
$G = \xi D$ — gain (unitless)
$G_{\text{dB}} = 10 \log G$
Half-wave: $D = 1.64, R_{\text{rad}} = 73 \Omega$
Hertzian: $D = 1.5$

FRIIS / LINK

P_t — tx power (W)
P_r — rx power (W)
G_t, G_r — linear gains
R — range (m)
$L_{\text{fs}} = (4\pi R/\lambda)^2$ — path loss
3 dB $\rightarrow$ 2; 10 dB $\rightarrow$ 10
$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$

RADAR

σ_t — RCS (m^2)
f_d — Doppler shift (Hz)
v_r — radial vel. (m/s)
$P_r \propto 1/R^4$ (monostatic)
$f_d = 2v_r/\lambda$ (round trip)
$c = 3 \times 10^8 \text{ m/s}$
$d > 0$: target approaching

TEXTBOOK REFERENCE TABLES

Table 7-1 — Lossy media: α , β , η_c , u_p , λ by regime

Table 7-1 Expressions for α , β , η_c , u_p , and λ for various types of media.

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu\epsilon}$	$\omega \sqrt{\mu\epsilon}$	$\sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$(1 + j) \frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω / β	$1 / \sqrt{\mu\epsilon}$	$1 / \sqrt{\mu\epsilon}$	$\sqrt{4\pi f / \mu \sigma}$	(m/s)
$\lambda =$	$2\pi / \beta = u_p / f$	u_p / f	u_p / f	u_p / f	(m)
Notes: $\epsilon' = \epsilon$; $\epsilon'' = \sigma / \omega$; in free space, $\epsilon = \epsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' = \sigma / \omega \epsilon < 0.01$ and a good conducting medium if $\epsilon''/\epsilon' > 100$.					

Table 7-2 — dB $\leftrightarrow$ linear power ratios

Table 7-2 Power ratios in natural numbers and in decibels.

G	G [dB]
10^x	$10x$ dB
4	6 dB
2	3 dB
1	0 dB
0.5	−3 dB
0.25	−6 dB
0.1	−10 dB
10^{-3}	−30 dB

Table 8-1 — Normal incidence plane wave $\leftrightarrow$ transmission line analogy

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]	Transmission Line [Fig. 8-4(b)]
$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^i (e^{-jk_1 z} + \Gamma e^{jk_1 z})$ (8.11a)	$\tilde{V}_1(z) = V_0^+ (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z})$ (8.11b)
$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} (e^{-jk_1 z} - \Gamma e^{jk_1 z})$ (8.12a)	$\tilde{I}_1(z) = \frac{V_0^+}{Z_{01}} (e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z})$ (8.12b)
$\tilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^i e^{-jk_2 z}$ (8.13a)	$\tilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2 z}$ (8.13b)
$\tilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E_0^i}{\eta_2} e^{-jk_2 z}$ (8.14a)	$\tilde{I}_2(z) = \tau \frac{V_0^+}{Z_{02}} e^{-j\beta_2 z}$ (8.14b)
$\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$	$\Gamma = (Z_{02} - Z_{01}) / (Z_{02} + Z_{01})$
$\tau = 1 + \Gamma$	$\tau = 1 + \Gamma$
$k_1 = \omega \sqrt{\mu_1 \epsilon_1}$, $k_2 = \omega \sqrt{\mu_2 \epsilon_2}$	$\beta_1 = \omega \sqrt{\mu_1 \epsilon_1}$, $\beta_2 = \omega \sqrt{\mu_2 \epsilon_2}$
$\eta_1 = \sqrt{\mu_1 / \epsilon_1}$, $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

Table 8-2 — Γ , τ , R , T for normal / $\perp$ / $\parallel$ polarization

Table 8-2 Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.			

Table 8-3 — TE / TM / TEM full field components in rectangular waveguide

Table 8-3 Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$, filled with a dielectric material with constitutive parameters ϵ and μ . The TEM case, shown for reference, pertains to plane-wave propagation in an unbounded medium.

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y / Z_{TE}$ $\tilde{H}_y = \tilde{E}_x / Z_{TE}$ $\tilde{H}_z = H_0 \cos \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y / Z_{TM}$ $\tilde{H}_y = \tilde{E}_x / Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = E_{x0} e^{-j\beta z}$ $\tilde{E}_y = E_{y0} e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y / \eta$ $\tilde{H}_y = \tilde{E}_x / \eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2}$ $\beta = k \sqrt{1 - (f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0} / \sqrt{1 - (f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega \sqrt{\mu\epsilon}$ $u_{p0} = 1 / \sqrt{\mu\epsilon}$